

Project Demo Planning

Date	4 July 2026
Team ID	xxxxxxxxxx
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Project Demo Planning:

A well-structured demo plan ensures that the project is presented effectively, covering all key aspects in a clear and organized manner. This document outlines the plan for demonstrating the OptiCrop project, including the flow of the demo and key features to highlight.

S.N O	Demo Section	Description	Duration (mins)	Responsible Member
1	Project Introduction	Introduce OptiCrop — what it does, why it matters, and the agricultural problem it solves	2 mins	A.Sowmya k.Geethika Md.Karishma J.Sravani N.Neha Neeharika
2	Dataset & Analysis	Show Crop_recommendation.csv dataset, explain features (N,P,K,temp,humidity,pH,rainfall) and demonstrate univariate & multivariate analysis plots	3 mins	A.Sowmya k.Geethika Md.Karishma J.Sravani N.Neha Neeharika
3	Data Preprocessing	Demonstrate null value check, outlier detection using boxplot, seasonal crop extraction, and train/test split	7 mins	A.Sowmya k.Geethika Md.Karishma J.Sravani N.Neha Neeharika
4	Model Building	Show K-Means Elbow method, Logistic Regression training, classification report with 94% accuracy, and model saving as model.pkl	5 mins	A.Sowmya k.Geethika Md.Karishma J.Sravani N.Neha

5	Flask App Demo	Run app.py, open browser, show Home page, About page, then enter test values (N=105,P=35,K=40,T=25,H=64,pH=7,R=160) and demonstrate crop prediction output	7 mins	A.Sowmya k.Geethika Md.Karishma J.Sravani N.Neha Neeharika
6	Conclusion & Summary	Summarize key achievements, model accuracy, features implemented, and future scope of OptiCrop	5 mins	A.Sowmya k.Geethika Md.Karishma J.Sravani N.Neha Neeharika

Demo Flow Summary:

Step	Activity	Notes
1	Introduction & Problem Statement	Explain agricultural crop selection problem and how OptiCrop solves it using ML
2	Solution Overview	Walk through the project architecture: Dataset → Preprocessing → ML Model → Flask App
3	Live Feature Demonstration	Run Jupyter notebook cells + Run app.py + Show prediction result in browser
4	Q&A; Session	Answer any questions about the ML model, accuracy, or Flask application functionality